

Basic SQL Server Health Check — Key Steps for Optimal Performance

When performing a **SQL Server health check**, follow these steps to ensure the environment is stable and performing optimally:

1. Verify SQL Services

- Check that all SQL Server–related services (Database Engine, SQL Agent, SSIS, etc.) are running.

2. Review Error Logs

- Examine SQL Server and Windows Event Logs for critical errors or recurring warnings.

3. Monitor Server Resources

- Analyze CPU utilization, memory usage, and disk I/O performance.
- Ensure there are no resource bottlenecks.

4. Check TempDB Health

- Verify adequate data files, proper sizing, and no contention issues (e.g., PFS, SGAM, or GAM latch waits).

5. Validate Database Status

- Confirm all databases are online and accessible.
- Check for suspect or recovery-pending states.

6. Backup & Restore Validation

- Ensure backups (full, differential, and log) are recent and restorable.
- Perform test restores periodically.

7. Review SQL Agent Jobs

- Verify success of critical jobs such as backups, index maintenance, and **DBCC CHECKDB** integrity checks.

8. Analyze Performance Metrics

- Review wait statistics, blocking sessions, and long-running queries.
- Capture and analyze performance trends.

9. Validate High Availability / DR

- If **Always On** or **Log Shipping** is configured:
 - Check synchronization status, replica health, and job execution results.

10. Review Security Settings

- Check for failed logins, orphaned users, and unnecessary privileges.
- Validate role-based access and login policies.

11. Check Configuration Parameters

- Ensure key settings like **Max Server Memory**, **MAXDOP**, and **Cost Threshold for Parallelism** are tuned properly.

12. Review Alerts & Monitoring Tools

- Validate alert configurations and ensure monitoring dashboards (e.g., SolarWinds, CloudWatch, or custom scripts) are reporting proactively.